POKHARA UNIVERSITY

Semester: Spring

Level: Bachelor
Programme: BE
Course: Microprocessors

Year : 2013
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

a) Explain Harvard architecture and give reason behind this architecture being faster than Von-Neumann architecture. 7
b) Define flags and addressing modes. What are the different addressing modes used in 8085 microprocessor? Explain with examples. 8
a) Draw the timing diagrams for the instruction IN 4FH. 8
b) Write a 8085 program to find the largest of three numbers. 7
a) Write an 8085 ALP to find ODD and EVEN number among six bytes of data: 23H, 41H, 56H, AFH, C5H & A7H stored in the memory location starting from 2500H and place the ODD and EVEN numbers in memory location starting from 2600H and 2400H respectively. 8
b) Design a circuit to interface a 8 KB ROM to 8085 microprocessor. The starting address being1000H. 7
a) Draw the functional block diagram of 8086 Microprocessor and explain how segmentation and pipelining concepts are implemented in this microprocessor. 8
b) Write an assembly language programming for 8086 to find the square root of a given number. Assume that a number is of two digits and is perfect square. 7
a) Specify the functions of the following 8086 instructions: 7
AAA, LDS, DIV, MOVSB, NOT, CALL, ROL
b) Define IVT. Explain interrupt vector table for 8086 microprocessor. 8
a) Explain the RS232C standard for serial communication. Explain its advantages and disadvantages. 7
b) Explain the functional block diagram of 8259 PIC. 8
7. Write short notes on: *(Any two)* 2×5
a) Synchronous and Asynchronous data bus
b) Modes of operation of 8255
c) Memory mapped I/O and I/O mapped I/O

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